Name :- Aarsh Singh

Roll no :- 22515000003

Theory Assignment - 3

Answers

1. Frontend JavaScript runs in the browser and handles webpage interaction and the user interface. Backend JavaScript runs on the server and handles databases, authentication, and server-side logic.

2. DOM (Document Object Model) is a tree-like representation of an HTML document. JavaScript uses it to access and change webpage elements, content, and styles.

3. An event in JavaScript is an action or occurrence such as a click, key press, mouse movement, or form submission that code can respond to.

4. The event loop manages asynchronous tasks. It checks when the call stack is empty and then moves ready callbacks from the task queue to the stack for execution.

5. Hoisting is JavaScript's behavior of processing declarations before code execution. Function declarations can be used before they appear; var is hoisted and initialized as undefined, while let and const are not usable before initialization.

6. Event bubbling is when an event starts on the target element and travels upward through its parent elements in the DOM.

7. A named function has a name and can be called by that name. An anonymous function has no name and is often assigned or passed as a callback. An arrow function uses => syntax and has lexical this.

8. A for-of loop iterates over values in an iterable, such as an array. A for-in loop iterates over enumerable property keys, usually object keys.

9. Five array methods: push() adds an item at the end; pop() removes the last item; shift() removes the first item; unshift() adds an item at the beginning; map() creates a new array by applying a function to each item.

10. A callback is a function passed to another function as an argument, to be called later, often after an event or when a task is completed.